



Special Cases

- Degeneracy
- Alternative Optima
- Unbounded LP Solution
- Infeasible LP

Reference Only



Degeneracy

$$\begin{aligned} \text{Max } & 3x_1 + 9x_2 \\ \text{subject to } & x_1 + 4x_2 \leq 8 \\ & x_1 + 2x_2 \leq 4 \\ & x_1, x_2 \geq 0 \end{aligned}$$

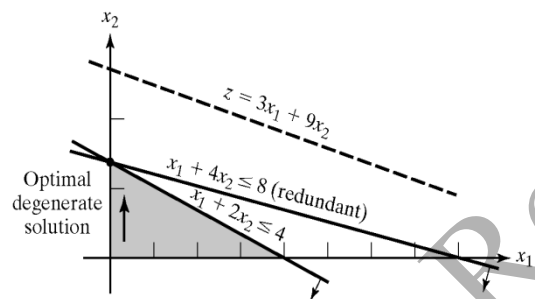


Figure 3.7
LP degeneracy in Example 3.5-1.

	x1	x2	x3	x4	Solution
z	-3	-9	0	0	0
x3	1	4	1	0	8
x4	1	2	0	1	4

	x1	x2	x3	x4	Solution
z	-3/4	0	9/4	0	18
x2	1/4	1	1/4	0	2
x4	1/2	0	-1/2	1	0

	x1	x2	x3	x4	Solution
z	0	0	3/2	3/2	18
X2	0	1	1/2	-1/2	2
x1	1	0	-1	2	0



Alternative Optima

$$\begin{aligned} \text{Max } & 2x_1 + 4x_2 \\ \text{subject to } & x_1 + 2x_2 \leq 5 \\ & x_1 + x_2 \leq 4 \\ & x_1, x_2 \geq 0 \end{aligned}$$

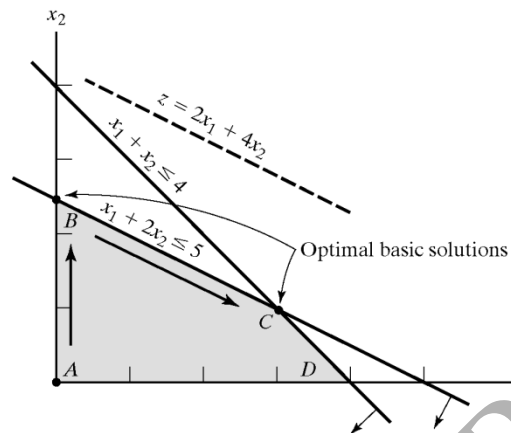


Figure 3.9
LP alternative optima in Example 3.5-2.

	x1	x2	x3	x4	Solution
z	-2	-4	0	0	0
x3	1	2	1	0	5
x4	1	1	0	1	4

	x1	x2	x3	x4	Solution
z	0	0	2	0	10
x2	1/2	1	1/2	0	5/2
x4	1/2	0	-1/2	1	3/2

	x1	x2	x3	x4	Solution
z	0	0	2	0	10
X2	0	1	1	-1	1
x1	1	0	-1	2	3



Unbounded LP Solution

$$\begin{aligned} &\text{Max } 2x_1 + x_2 \\ &\text{subject to} \\ &x_1 - x_2 \leq 10 \\ &2x_1 \leq 40 \\ &x_1, x_2 \geq 0 \end{aligned}$$

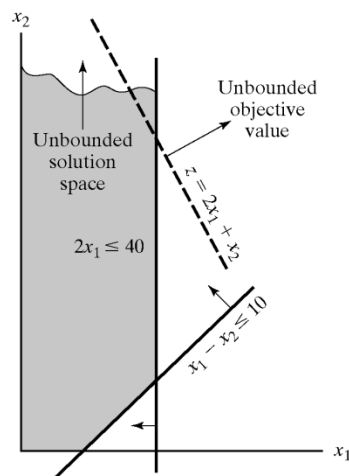


Figure 3.10

LP unbounded solution in Example 3.5-3.

	x1	x2	x3	x4	Solution
z	-2	-1	0	0	0
x3	1	-1	1	0	10
x4	2	0	0	1	40



Status of an LP

	Optimal	Unbounded	Infeasible
LP	Yes	No	No

	Optimal	Unbounded	Infeasible
LP	No	Yes	No

	Optimal	Unbounded	Infeasible
LP	No	No	Yes